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## Rotational Lock Mounting Systems and Methods of Use Thereof

### Abstract

A rotational lock mounting system for use with photovoltaic (PV) modules (e.g., panels) and arrays. The mounting system includes an anchor base and a standoff. The standoff further includes a standoff mounting portion and a locking component coupled thereto. The mounting system can be used by inserting the locking component of the standoff into the anchor base and rotating the standoff to secure the locking component within the anchor base.

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### Background/Summary

## TECHNICAL FIELD OF INVENTION

[0001] This invention generally relates to rotational lock mounting systems for photovoltaic (PV) modules (e.g., panels) and arrays and methods of use thereof.

## BACKGROUND OF THE INVENTION

[0002] A photovoltaic (PV) installation typically includes a collection of photovoltaic modules combined and secured to a support structure that combines each of the photovoltaic components to form a photovoltaic array. Typically, photovoltaic arrays are placed in an outdoor location, commonly rooftops, so that the photovoltaic arrays are exposed to sunlight in order to produce electricity. The manner in which the PV arrays are attached to surfaces varies.

## SUMMARY OF THE INVENTION

[0003] In an aspect, the present disclosure relates to a rotational lock mounting system. Such systems can be used for mounting photovoltaic modules onto a surface. The rotational lock mounting system includes an anchor base and a standoff.

[0004] In an aspect, the present disclosure relates to an anchor base component. Anchor bases include at least two side walls that extend perpendicularly from a horizontal mounting portion. The side walls include at least a pair of flanges. A first pair of flanges can be positioned at the portion of the side walls furthest away from the horizontal mounting portion (i.e. the side wall upper portion). A second pair of flanges can be positioned closer to the horizontal mounting portion (i.e. the side wall bottom portion) than the first pair of flanges. This forms a channel capable of receiving a portion of the standoff. Pairs of flanges extend from the side walls towards the channel formed within, forming a groove capable of receiving a component of the standoff within the channel. Additionally, the first pair of flanges can be substantially longer than the second pair of flanges.

[0005] In an aspect, the present disclosure relates to a standoff component. The standoff includes a standoff mounting portion that couples photovoltaic (PV) components. The standoff mounting portion may be substantially rod-like in shape, though other shapes are possible.

[0006] The standoff further includes a locking component coupled to the standoff mounting portion and capable of coupling to the anchor base. The locking component includes a first narrow portion, a middle portion, and a second narrow portion. The two narrow portions are at opposing ends of the locking component. The middle portion is also wider than the two narrow portions. The middle portion includes at least two arms extending from the middle portion in parallel directions. Such arms each include at least one nub on their ends. The nubs extend away from the center of the locking component. A flange can be located at the border of the mounting portion and the locking component.

[0007] The locking component can further include a stabilizing portion located on the end of the locking component opposing the standoff mounting portion. The stabilizing portion can include a deformable pad. This pad is configured to be placed adjacent to a flat surface of the horizontal mounting portion of the anchor base, providing the rotational lock mounting system additional stability. The pad also assists in ensuring that a standoff is rotated to the correct position when locking the standoff in an anchor base. The horizontal mounting portion can include a raised portion or a flat portion that the deformable pad can abut.

[0008] In an aspect, the present disclosure relates to methods of using rotational lock mounting systems. Such methods include using such mounting systems to mount a photovoltaic module.

[0009] In an exemplary method, a standoff can be inserted into an anchor base such that the bottom surface of the locking component abuts an internal surface of the anchor base. As such, the deformable pad may abut the horizontal mounting portion. After insertion, the arms of the locking component run parallel to the anchor base channel. The standoff can then be rotated at least a quarter turn. Once rotated, surfaces of the arms of the locking component abut surfaces of the channel. Nubs may abut flanges once the standoff is rotated. Also after rotation, the flange of the standoff and the top surfaces of the arms of the locking component abut the first pair of flanges of

the side walls.
[0010] These and other objects and advantages of the invention will become apparent from the following detailed description of the invention. Both the foregoing general description and the following detailed description are exemplary and explanatory only and are intended to provide further explanation of the invention as claimed.
[0011] The accompanying drawings are included to provide a further understanding of the invention and are incorporated in and constitute part of this specification, as well as illustrate several embodiments of the invention that together with the description serve to explain the principles of the invention.

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## Description

BRIEF DESCRIPTION OF DRAWINGS
[0012] The features and components of the following figures are illustrated to emphasize the general principles of the present disclosure. Corresponding features and components throughout the figures can be designated by matching reference characters for the sake of consistency and clarity.
[0013] FIG. **1** displays a front, planar view of aspects of the present disclosure including an anchor base and a standoff.
[0014] FIG. **2** displays a front, planar view of aspects of the present disclosure including an anchor base interacting with a standoff.
[0015] FIG. **3** displays a front, planar view of aspects of the present disclosure including an anchor base interacting with a rotated standoff.
[0016] FIG. **4** displays a non-planar top view of aspects of the present disclosure including an anchor base and a standoff.
[0017] FIG. **5** displays a front, planar view of aspects of the present disclosure including an anchor base interacting with a rotated standoff.
[0018] FIG. **6** displays a front, planar view of aspects of the present disclosure including an anchor base interacting with a standoff.
[0019] FIG. **7** displays a front view of aspects of the present disclosure including an anchor base and a locking component.
[0020] FIG. **8** displays a view of a securing cover with an anchor base and a locking component according to an aspect of the present invention.
DETAILED DESCRIPTION OF THE INVENTION
[0021] Embodiments of the invention will be described more fully hereinafter with reference to the accompanying drawings, in which embodiments of the invention are shown. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art.
[0022] [In the following description, numerous specific details are set forth. However, it is to be understood that embodiments of the invention may be practiced without these specific details. In other instances, well-known methods, structures, and techniques have been shown in detail in order not to obscure an understanding of this description.
[0023] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. Any systems and components similar or equivalent to those described herein can be used in the practice or testing of the present invention. All publications mentioned are incorporated herein by reference in their entirety.
[0024] The use of the terms “a,” “an,” “the,” and similar referents in the context of describing the presently claimed invention (especially in the context of the claims) are to be construed to cover

both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context.

[0025] Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein.

[0026] Use of the term “about” is intended to describe values either above or below the stated value in a range of approx. +/−10%; in other embodiments the values may range in value either above or below the stated value in a range of approx. +/−5%; in other embodiments the values may range in value either above or below the stated value in a range of approx. +/−2%; in other embodiments the values may range in value either above or below the stated value in a range of approx. +/−1%. The preceding ranges are intended to be made clear by context, and no further limitation is implied. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate the invention and does not pose a limitation on the scope of the invention unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the invention.

[0027] The present disclosure can be understood more readily by reference to the following detailed description, examples, drawings, and claims, and their previous and following description. However, before the present systems and components are disclosed and described, it is to be understood that this disclosure is not limited to the specific systems and components disclosed unless otherwise specified, as such can, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting.

I. Rotational Lock Mounting Systems

[0028] In an aspect, the present disclosure relates to a rotational lock mounting system **10**. The rotational lock mounting system **10** can be used in combination with solar photovoltaic (PV) mounting systems. Rotational lock mounting systems **10** can be used to mount PV modules (e.g., panels) and arrays to a variety of surfaces including roofs. Roofs include but are not limited to tile roofs, shingle roofs, rolled asphalt roofs, wooden roofs, roofs comprising structural decking, roofs comprising rafters, and any other roof known in the art. Such a system **10** incorporates rotational lock fastening into PV mounting systems in a novel mannert. This provides faster and simpler installation of PV components, as described herein. Such rotational lock mounting systems **10** increase space between a surface and PV components mounted to such systems **10**. PV components include a variety of PV components such as PV modules, PV arrays, PV apparatus, PV subsystems, and any other similar components known in the art.

[0029] In an aspect, a system **10** includes anchor bases **100**, standoffs **200**, and in some instances a cover **300**, as shown in FIGS. **1-8**. Anchor bases **100** are configured to rest on a surface while standoffs **200** couple to PV components. The combination of the anchor bases **100** and standoffs **200** may be used on many different surfaces, including many different roof types and angles. Anchor bases **100** and standoffs **200** are configured to interact with one another to provide a unique interlocking function that allows a simple, fast, and secure assembly thereof. The anchor bases **100** and standoffs **200** are capable of being produced in various sizes. If the size of one component (i.e. an anchor base **100** or a standoff **200**) is altered, the other component will likely also have to be adjusted such that the two components **100** and **200** remain compatible. Sizes may be altered based on the dimensions, including, but not limited to height, width, weight, and the like, of tile types (e.g., W or T tiles) and PV components to be used in conjunction with one or more anchor bases **100** and standoffs **200** of a rotational lock mounting system **10**. Additional discussions of rotational lock mounting system **10** dimensions are discussed further herein. [0030] A. Anchor Bases and

Components Thereof

[0031] In an aspect, rotational lock mounting systems **10** include at least one anchor base **100**. Anchor bases **100** may include a variety of shapes including, but not limited to, substantially quadrilateral (see FIGS. **1-3**), circular (FIGS. **4-6**), oblong (FIG. **8**), and other shapes. In some aspects, the base **100** can be the same shape as the tiles on the roof on which is attached. In any case, the anchor base **100** can include side walls **110**, as shown in FIGS. **1-6**. In an exemplary aspect, an anchor base **100** may include two, three, four, or more side walls **110**. In an additional, exemplary aspect, an anchor base **100** may include two side walls **110**. Such side walls **110** include upper/top portions **112** extending upward from lower/bottom portions **114**, as shown in FIGS. _. Side walls further include a first/outer surface **116** and a second/inner surface **118**. Side walls **110** extend perpendicularly from a horizontal mounting portion **150**, as discussed further below.

[0032] In an aspect, side walls **110** form a channel **120**, as shown in FIGS. **1-2** and **7**. The channel **120** is configured to receive the standoff **200**, discussed below. The heights of the side walls **110** can vary, and may include heights of about 0.1 inches up to about 0.25 inches, about 0.125 inches up to about 0.225 inches, or about 0.15 inches up to about 0.20 inches. Higher side walls **110** may form deeper channels **120** configured to receive larger components associated with standoffs **200**, as discussed further herein. Anchor bases **100** may also include widths. Such widths may be determined by the spacing of side walls **110**, as shown in FIG. **1**. Example widths include but are not limited to about 0.5 inches, about 1 inch, about 1.5 inches, about 2 inches, about 2.5 inches, about 3 inches, about 3.5 inches, about 4 inches, about 4.5 inches, about 5 inches, about 5.5 inches, about 6 inches, about 6.5 inches, about 7 inches, about 7.5 inches, about 8 inches, about 8.5 inches, about 9 inches, about 9.5 inches, or about 10 inches. Anchor bases **100** may also include lengths. Example lengths include but are not limited to about 0.5 inches, about 1 inch, about 1.5 inches, about 2 inches, about 2.5 inches, about 3 inches, about 3.5 inches, about 4 inches, about 4.5 inches, about 5 inches, about 5.5 inches, about 6 inches, about 6.5 inches, about 7 inches, about 7.5 inches, about 8 inches, about 8.5 inches, about 9 inches, about 9.5 inches, or about 10 inches. In some examples, an anchor base **100** may be circular in shape (See FIGS. **4-6**). In such examples, a diameter of an anchor base **100** may be used to describe the anchor base **100**. In such examples, a diameter of an anchor base **100** may include but is not limited to a diameter of about 0.5 inches, about 1 inch, about 1.5 inches, about 2 inches, about 2.5 inches, about 3 inches, about 3.5 inches, about 4 inches, about 4.5 inches, about 5 inches, about 5.5 inches, or about 6 inches.

[0033] The upper portions **112** of side walls **110** may additionally include a pair of flanges **122**, as shown in FIGS. **1-6**. Such flanges **122** extend away from the second surface **118** of the side wall **110** and towards the middle of the channel **120**. In such an example, the flanges **122** on two side walls **110** may extend towards each other, as shown in FIG. **1**. The flanges **122** may include lengths. Lengths may range from about 0.10 inches up to about 0.50 inches, or about 0.20 inches up to about 0.40 inches. In a particular example, upper flanges may be about 0.20 inches in length.

[0034] As a non-limiting example, as shown in FIGS. **1-3**, side walls **110** may include at least two pairs of flanges **122**, **126**. In such an example, a side wall **110** may include upper flanges **122** with a second pair of flanges **126** underneath. The secondary flanges **126** extend in the same direction as the upper flange **122** (i.e. towards the channel **120** interior). As a non-limiting example, the secondary flanges **126** may be substantially shorter than the upper flanges **122**. In such an example, the secondary flanges **126** do not extend as far into the interior of the channel **120** as do the upper flanges **122**. In such an aspect, the upper flanges **122** and the secondary flanges **126** form a small channel or groove **130** between themselves along the upper portions **112** of the side walls **110**, as shown in FIGS. **1-3**.

[0035] In an aspect, anchor bases **100** further include horizontal mounting portions **150**, as shown in FIGS. **1-6** and discussed herein. Such horizontal mounting portions **150** are configured to abut a surface **20** upon which the system **10** rests. Horizontal mounting portions **150** include an upper/top surface **152** and an opposing, bottom surface **154**, as shown in FIG. **3**. In an aspect, the horizontal mounting portions **150** further include outer arms **156** extending in a horizontal direction from the outside surface **116** at the bottom of the side walls. As a non-limiting example, a horizontal mounting portion **150** can be configured to include a substantially flat upper surface **152**, as shown in FIG. **1**. As an additional, non-limiting example, a horizontal mounting portion **150** can be configured to include a raised portion on its upper surface **152** (i.e. non-flat/textured), as shown in FIG. **4**. Horizontal mounting portions **150** additionally include a plurality of apertures **156**, as shown in FIGS. **6-8**. Such apertures **158** receive fastening devices **160** and allow a rotational lock mounting system **10** to be coupled to a surface. In a non-limiting example, an aperture **158** of a horizontal mounting portion **150** may receive a screw/fastener **160** that couples the system **10** to a surface **20**, as shown in FIG. **7**. In additional examples, butyl padding **170** may be placed below a locking system **10** before it is coupled to a surface, preventing water leaking through the surface **20** following coupling.

Standoffs and Components Thereof

[0036] In an aspect, the present disclosure relates to a standoff **200**. The standoff **200** includes an upper end **202**, a lower end **204**, and a middle section **206**. A mounting portion **210** can be found at the upper end **202**. The mounting portion **210** can be configured to couple to a PV module (e.g., panel) or array (not shown). A mounting portion **210** may include an aperture **212** extending through the mounting portion **210** configured to assist in coupling a PV module to the standoff **200**, as shown in FIGS. **4-8**. The mounting portion **210** can be configured to be substantially rod-like. In such an example, a mounting portion **210** may have a diameter of about 0.25 inches up to about 2 inches. In additional examples, a mounting portion **210** may have a diameter of about 0.5 inches up to about 1.75 inches, of about 0.75 inches up to about 1.5 inches, or about 1.0 inches up to about 1.25 inches. In an example, a mounting portion **210** may also include a height from about 2 inches up to about 8 inches. In additional examples, a mounting portion **210** may include heights from about 3 inches up to about 7 inches, or about 4 inches up to about 6 inches. The mounting portion **210** may additionally include other shapes including but not limited to prism shapes of various polygons. In an aspect, a standoff **200** also includes a flange **220**. Such a flange **220** can be located along the middle section **206** of the standoff **200**. In such an aspect, a flange **220** can be located at the border of a standoff mounting portion **210** and a locking component **250**, as discussed further below.

[0037] In an aspect, a standoff **200** includes the locking component **250** found at a bottom end **204** of the standoff **200**. The locking component **250** includes a first narrow portion **252** adjacent the flange **220** of the mounting portion **210**, as shown in FIG. **1**. In an aspect, the first narrow portion **252** has a width/diameter that is smaller than the distance between the flanges **122** of the base **100**. Such a configuration allows the locking component **250** to rest within the channel **120** formed by side walls **110**. As discussed above, larger locking components **250** may require varied dimensions of anchor bases **100**. In an example, a wider locking component **250** may require walls **110** of an anchor base **100** to be spaced further apart. In an additional example, a taller locking component **250** may require walls **110** of an anchor base **100** to be include larger heights. In such examples, larger locking components **250** may be necessitated by larger or varied PV components to be mounted in conjunction with a rotational lock mounting system **10**.

[0038] In such aspect, a middle portion **260** of the locking component **250** extends from the first narrow portion **252**, as shown in FIGS. **1-6**. The middle portion **260** may be wider than the first narrow portion **252**. Extending from opposite sides of the middle portion **260** are at least two arms **262**, best shown in FIGS. **3** and **6**. The middle portion **260** includes an upper surface **268** and side surfaces **270** running along the arms **262**. The side surfaces **270** may be substantially flat. In an additional aspect, the arms **262** may form a slightly curved shape, as shown in FIG. **2**. In an aspect, the arms **262** may vary in width along their length.

[0039] Each of the at least two arms **262** further include nubs **266** on the ends **264** of the arms **262**. In such an aspect, a nub **266** on the end **264** of an arm **262** may abut the secondary flanges **126** when a locking component **250** is rotated within an anchor base **100** as shown in FIG. **3**, or the inner surfaces **118** of the side walls **110** of the base **100** under the flanges **122** as shown in FIG. **6**. Such engagement assists in further securing the locking component **250** within the anchor base **100**, as further discussed herein.

[0040] In an exemplary aspect, the length of the middle portion **260** running from a first end **264** of a first arm **262** to a second end **264** of a second arm **262** equals approximately the distance between the secondary flanges **126** as shown in FIG. **3**. In another exemplary aspect, the length of the middle portion **260** running from a first end **264** of a first arm **262** to a second end **264** of a second arm **262** equals approximately the distance between the inner surfaces **118** of the side walls **110**, as shown in FIG. **6**. In another aspect, the distance from a first nub **266** on a first end **264** to a second nub **266** on a second end **264** of arms **262** is larger than the length from one arm end **264** to the other arm end **264**, creating a friction hold of the locking component **250** within the channel **120** of the side walls **110**.

[0041] In an aspect, a second narrow portion **280** further extends from the middle portion **260**, as shown in FIG. **3**. This second narrow portion **280** serves as a stabilizing portion **280** and is referred to interchangeably as such herein. The stabilizing portion **280** is used to engage the upper surface **152** of the mounting portion **150** of the anchor base **100**. In an aspect, the stabilizing portion **280** may include an integrated deformable pad **282**. A deformable pad **282** may be made of any material capable of deformation including a variety of polymers. Applicable polymers include but are not limited to ethylene propylene diene monomers (EPDMs). The deformable pad **282** can be pressed up against an upper surface **152** of the horizontal mounting portion **150**. The combination of the stabilizing portion **280** and the deformable pad **282** assist in stabilizing the locking component **250** within the channel **120**. As such, the deformable pad **282** firmly abuts the upper surface **152** to stabilize the locking component.

Securing Cover

[0042] As shown in FIG. **8**, the rotational lock mounting system **10** can include a securing cover **300** configured to work with the anchor base **100** and standoff **200**, as shown in FIG. **8**. In an aspect, the securing cover **300** includes a horizontal base **310**. The horizontal base **310** includes an aperture **312** configured to receive the upper portion **210** of the standoff **200**. The horizontal base **310** includes sides **320**, **322**, **324**, **326**. A pair of short clamps **330** extend downwardly from sides **320**, **324** opposite one another. Similarly, a pair of long clamps **340** extend downwardly from sides **322**, **326** opposite one another. In an aspect, the short clamps **330** are configured to engage with upper surface of the outer arms **152** of the horizontal mounting portion **150** of the anchor base **100**, whereas the long clamps **340** are configured to come in contact with the surface **20** on which the base **100** is secured.

[0043] In an aspect, the short clamps **330** include a top end **332** and a bottom end **334**, with the top end **332** being connected to the sides **320**, **324** of the horizontal base **310**. Flanges **336** can extend from the bottom end **334** in an outward horizontal direction. The flanges **336** of the short clamps **330** are configured to interact with/abut the outer arms **156** of the horizontal mounting portions **150**. In an aspect, the flanges **336** include apertures **338**. These apertures **338** can be configured to be aligned with the apertures **158** of the horizontal mounting portion **150**. In such instances, the apertures **338** can be configured to receive the same fasteners **160** used to secure the anchor base **100** to the surface **20**. In another instance, the apertures **338** can be sized to fit over the fasteners **160** of the anchor base **100** after being secured.

[0044] In an aspect, the long clamps **340** include a top end **342** and a bottom end **344**, with the top end **342** connecting to the sides **322**, **326** of the horizontal base **310**. Flanges **346** can extend from the bottom end **344** of the long clamp **340** in an outward horizontal direction. The long clamps **340** can be configured to come into contact with side portions of the horizontal mounting portion **150** at the bottom ends **344**, and the flanges **346** can come in contact or come close to the surface **20** on which the anchor base **100** is secured.

[0045] The securing cover **300** provides two functions—further securing the standoff **200** to the anchor base **100**, as well as shielding portions of the anchor base **100** and the standoff **200** for unnecessary exposure of the elements.

II. Methods of Use

[0046] In an aspect, the present disclosure relates to methods of use of embodiments described herein. In such an aspect, a standoff **200** is configured to interact and couple with an anchor base **100**. Such interaction includes a portion of the standoff **200**, such as a locking component **250**, being inserted within a recess of an anchor base **100**. The locking component **250** can then be manipulated to secure the overall standoff **200** within the anchor base **100**, as further described below.

[0047] As a non-limiting example, a standoff **200** can be secured to an anchor base **100** by first inserting the locking component **250** into a channel **120** of the base **100**, as shown in FIGS. **1-2**. In such example, the locking component **250** is inserted until the stabilizing portion **280** firmly abuts the upper surface **152** of the horizontal mounting portion **150**. In an aspect where a deformable pad **282** is coupled to the stabilizing portion **280**, the deformable pad **282** will firmly abut the upper surface **152** upon insertion. After such insertion, the locking component **250** is oriented such that arms **262** of the middle portion **260** run parallel with the channel **120** of the base **100**, as shown in FIG. **2**.

[0048] In such an example, a locking component **250** can be manipulated after insertion to secure the overall standoff **200** to the anchor base **100**. As such, the standoff **200** is then rotated until the arms **262** of the middle portion **260** abut an inner surface of the channel **120**. The standoff **200** can be rotated at least a quarter turn, at least a half turn, or at least a full turn (i.e. at least 90-degrees, at least 180-degrees, or at least 360-degrees). In a particular aspect, the standoff **200** can be rotated at least a quarter turn (i.e. at least 90-degrees). The inner surface that the arms **262** abut after rotation include the secondary flanges **126**, as shown in FIGS. **3** and **5**. After rotation, the flange **220** of the middle section **206** of the standoff **200** and the upper surface **268** of the arms **262** of the middle portion **260** are located substantially adjacent to the upper flanges **122** of the base **100**, as shown in FIGS. **3** and **4**. In an exemplary aspect, the flange **220** and the upper surface **268** of the arms **262** may firmly engage the upper flanges **122**. Such engagement adds stability to the standoff received within the anchor base. In such an example, ends **264** of the arms **262** additionally abut secondary flanges **126** after rotation, as shown in FIG. **3**.

[0049] In such an example, nubs **266** on the ends **264** of arms **262** are positioned within the channel **120** upon rotation. After rotation, nubs **266** may abut one or more of the lower flanges **126**, as shown in FIG. **7**. This interaction between nubs **266** and the lower flanges **126** may increase stability of the locking component **250** in its rotated configuration within the channel **120**. Once received, an audible clicking may occur in situations where a deformable pad **282** is not used in conjunction with a locking component **250** (not shown) where the clicking is caused by nubs **266** abutting lower flanges **126**. This auditory signal alerts the installer of reaching the locked position. In additional aspects, nubs **266** may rest against the stair-step configuration of flanges **122**, **126**, as shown in FIG. **5**. In such an aspect, nubs **266** may provide an auditory signal to inform an installer of reaching the locked position.

[0050] In an aspect, the deformable pad **282** additionally stabilizes the standoff **200** within the channel **120**, locking the standoff **200** in place. After rotation, the deformable pad **282** provides stability by exerting upward physical force on the locking component **250** as to hold it within the channel **120** by pressing upper surfaces **268** of arms **262** more firmly against upper flanges **122**, as shown in FIG. **3**.

[0051] In an aspect, installers may not hear a clicking sound, as described above, when a deformable pad **282** is used in conjunction with a locking component **250**, as shown in FIG. **4**. In such an aspect, an installer will be partly informed that a locking component **250** has been sufficiently rotated as to lock the standoff **200** into the anchor base **100** by one or more sensations.

Sensations include a feeling that the standoff **200** can be rotated no further or that rotation has become more difficult, as constrained by the morphology and interactions between portions of a locking component **250** and an anchor base **100**. One such interaction may occur as a result of the deformable pad **282**. A deformable pad **282** may be limited in the extent to which it can deform or may become more difficult to deform as a standoff **200** is rotated.

[0052] In an example, a polymeric deformable pad **282** may be limited by its molecular structure and characteristics as to how far it can rotate. In such aspects, a deformable pad **282** may offer increased resistance to rotational forces as an installer rotates a standoff **200** within an anchor base **100**. This may create a sensation informing the installer that the standoff **200** has been sufficiently rotated to achieve a lock. This resistance provided by a deformable pad **282** results from the interaction and friction between the deformable pad **282** and the upper surface **152** of the horizontal mounting portion **150**, as shown in FIG. **3**. In such aspects, deformation of a deformable pad **282** and positioning ends **264** of arms **262**, as well as nubs **266**, adjacent secondary flanges **126** of anchor bases **100** assist in locking a standoff **200** into an anchor base **100**.

[0053] An installer may also have a general understanding that the standoff may be rotated approximately 90-degrees to secure a lock, as shown in FIGS. **2-3** and **5-6**. As such, an installer may know, after rotating a standoff **200** 90-degrees that a lock has been formed.

[0054] In aspects in which a securable cover **300** is used, the standoff **200** can be inserted via the aperture **312** of the horizontal base **310** of the securable cover **300** prior to installation of the anchor base **100** and standoff **200**. The flanges **336** of the short clamps **330** can be aligned along the outer arms **156** of the horizontal mounting portion **150**, with the bottom portion **344** of the long clamps **340** aligned/adjacent the sides of the horizontal mounting portion **150**. From here, fasteners **160** can be inserted via the apertures **158**/**338** to secure both the cover **300** and the anchor base **100** to the surface **20**. Once fastened, the standoff **200** can be rotated in the manner discussed above.

[0055] Having thus described exemplary embodiments of the present invention, it should be noted by those skilled in the art that the disclosures are exemplary only and that various other alternatives, adaptations, and modifications may be made within the scope of the present invention. Accordingly, the present invention is not limited to the specific embodiments as illustrated herein, but is only limited by the following claims.

## Claims

**1**. A rotational lock mounting system for mounting photovoltaic modules onto a surface, the rotational lock mounting system comprising: a. an anchor base, wherein the anchor base comprises at least two side walls extending perpendicularly from a horizontal mounting portion and forming a channel to receive an additional component of the rotational lock mounting system; and b. a standoff, wherein the standoff comprises a standoff mounting portion configured to couple to at least one component of a photovoltaic module and a locking component opposing the standoff mounting portion configured to couple to the anchor base.

**2**. The rotational lock mounting system of claim 1, wherein the two side walls further comprise at least one pair of flanges positioned at the portion of the side walls furthest away from the horizontal mounting portion, the pair of flanges extending from the side walls towards the channel formed within.

**3**. The rotational lock mounting system of claim 1, wherein the standoff mounting portion comprises a flange located at the border of the standoff mounting portion and the locking component, and wherein a surface of the locking component abuts a surface of the flange.

**4**. The rotational lock mounting system of claim 1, wherein the standoff mounting portion extending away from the flange is substantially rod-like in shape.

**5**. The rotational lock mounting system of claim 1, wherein the locking component further comprises a first narrow portion, a middle portion, and a second narrow portion, wherein the two narrow portions are at opposing ends of the locking component, and wherein the middle portion is wider than the two narrow portions.

**6**. The rotational lock mounting system of claim 5, wherein the middle portion comprises at least two arms extending from the middle portion in parallel directions.

**7**. The rotational lock mounting system of claim 6, wherein the at least two arms each comprise a nub on the end of each of the at least two arms, and wherein the nub on the end of each of the at least two arms extends away from the center of the locking component.

**8**. The rotational lock mounting system of claim 1, wherein the locking component further comprises a stabilizing portion located on the end of the locking component opposing the standoff mounting portion.

**9**. The rotational lock mounting system of claim 8, wherein the stabilizing portion further comprises a deformable pad to be placed adjacent to a flat surface of the horizontal mounting portion of the anchor base to provide the rotational lock mounting system additional stability and to be engaged in locking the standoff into the anchor base.

**10**. The rotational lock mounting system of claim 8, wherein the horizontal mounting portion further comprises a raised portion, and wherein the stabilizing portion further comprises a deformable pad to be placed adjacent to the raised portion of the horizontal mounting portion of the anchor base to provide the rotational lock mounting system additional stability.

**11**. The rotational lock mounting system of claim 1, further comprising a securing cover configured to fit over the standoff and connect to the anchor base.

**12**. A method of mounting a photovoltaic module to a rotational lock mounting system, the method comprising: a. inserting a first component of the rotational lock mounting system into a second component of the same mounting system such that a bottom surface of the first component abuts an internal surface of the second component; and b. rotating the first component within the second component until the first surface is secure within the second component.

**13**. The method of claim 12, wherein the first component is a standoff, wherein the standoff comprises: a. a standoff mounting portion configured to couple to at least one component of a photovoltaic module; b. a flange positioned at an end of the standoff mounting portion; and c. a locking component opposing the standoff mounting portion, the locking component comprising: i. a first narrow portion; ii. a middle portion comprising at least two arms extending from the narrow portion in parallel directions, wherein the at least two arms each comprise a nub on the end of each of the at least two arms; iii. a second narrow portion; and iv. a stabilizing portion located on the end of the locking component opposing the standoff mounting portion.

**14**. The method of claim 13, wherein the second component is an anchor base, wherein the anchor base comprises: a. a horizontal mounting portion; and b. at least two side walls extending perpendicularly from the horizontal mounting portion and forming a channel, wherein the at least two side walls further comprise a pair of flanges within the channel.

**15**. The method of claim 14, wherein the standoff is first inserted into the anchor base such that the at least two arms of the middle portion extend in parallel to the length of the channel formed by the side walls.

**16**. The method of claim 15, wherein the standoff is then rotated by approximately 90-degrees to secure the locking component within the anchor base.

**17**. The method of claim 15, wherein the standoff is then rotated by approximately 45-degrees to secure the locking component within the anchor base.

**18**. The method of claim 16, wherein the nubs on the ends of each of the at least two arms firmly abut an inner surface of the side walls after the standoff is rotated 90-degrees to assist in securing the locking component within the anchor base.

**19**. The method of claim 18, wherein the at least two arms of the locking component abut the pairs of flanges positioned further within the channel after the standoff is rotated.

**20**. The method of claim 18, wherein the pair of flanges positioned at the end of the standoff mounting portion abuts a top surface of the two side walls.